Rex Chong

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EDUCATION

University of British Columbia *Expected May 2028*
B.A.Sc., Electrical Engineering | GPA: 89.2%

TECHNICAL SKILLS

Programming & HDL: C, Java/JavaFX, 8051 Assembly, RISC-V Assembly, SystemVerilog
Embedded & Hardware: PIC32, STM32, DE10-Lite FPGA, ADC, UART, I2C, PWM, interrupts, FSMs, Analog/Digital Circuits, Bluetooth
Tools: Quartus, QuestaSim, MATLAB/Simulink, Autodesk Fusion 360, Oscilloscope

TECHNICAL PROJECTS

Programmable Reflow Oven Controller *Feb 2026*

- Converted a toaster oven into a programmable reflow system; implemented an 8051-assembly FSM for preheat, soak, reflow and cooling profiles.
- Built thermocouple/ADC acquisition with cold-junction compensation for SSR-based PWM heater control (±1 °C regulation).
- Developed user interface with LCD display, speaker alerts, and fault detection logic. Along with live telemetry over UART for PC-side strip-chart plotting.
- Integrated IR sensor and motor for automated actuation and system sequencing as well as automatic process abortion if safety checkpoints are not met.

Magnetic Field Line-Following Robot *April 2026*

- Built a PIC32 controller and STM32 robot with inductive-coil magnetic sensing and intersection-based autonomous navigation; added joystick/PWM manual control and path retracing support.
- Designed controller user interface with LCD display screen for path and speed selection. Also implemented LED and audio feedback for car status and path choice.
- Implemented two-way Bluetooth interface to stream real time telemetry for integrated battery feedback and path progress, and CSV custom path uploading.
- Integrated object detection and automated claw mechanism using IR-triggered claw pickup/drop-off.
- Added safety features in collision avoidance via a VL53L0X time-of-flight sensor over I2C.

ARC4 Hardware Key-Cracking Engine *October 2026*

- Designed and verified a hardware-accelerated ARC4 stream cipher key-cracking engine in SystemVerilog for the Intel DE10-Lite FPGA.
- Architected hierarchical FSM controllers for KSA initialization and PRGA decryption phases, managing embedded block RAM (SRAM) read/write timing to prevent data hazards during array swaps.
- Implemented key-space search logic to iterate candidate keys against target signatures; verified control signals in QuestaSim prior to synthesis and mapped status outputs to onboard 7-segment displays and LEDs.

EXPERIENCE & LEADERSHIP

Math Tutor Network | Junior Academic Tutor *Sep 2025–Present*

- Provide weekly precalculus tutoring for high-school students, adapting explanations to their learning needs.
- Independently coordinate lesson planning, client scheduling and payment collection.
- Received returning bookings from clients across semesters.

Hope Lutheran Church | Youth Leader *Feb 2022–Present*

- Helped organize events for 20–40 children alongside a team of 5 leaders.
- Managed a weekly concession stand, keeping detailed records of sales numbers and supply costs to balance keeping prices cheap while contributing to the Youth Group budget.

AWARDS & ACHIEVEMENTS

Trek Excellence Scholarship (2026) | **Charles and Jane Banks Scholarship** (2026) | Dean's Scholar (2026) | Dean's List (2025) | BC Achievement Scholarship (2024) | Governor General's Bronze Medallion (2024) | RCM Gold Medal (2023 and 2019)